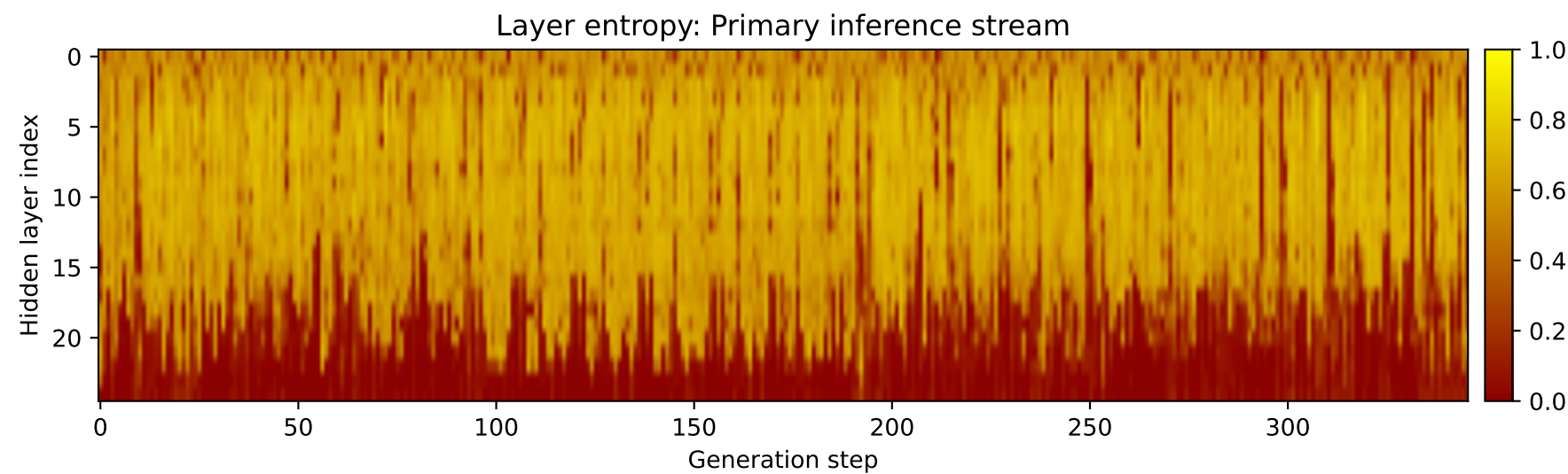
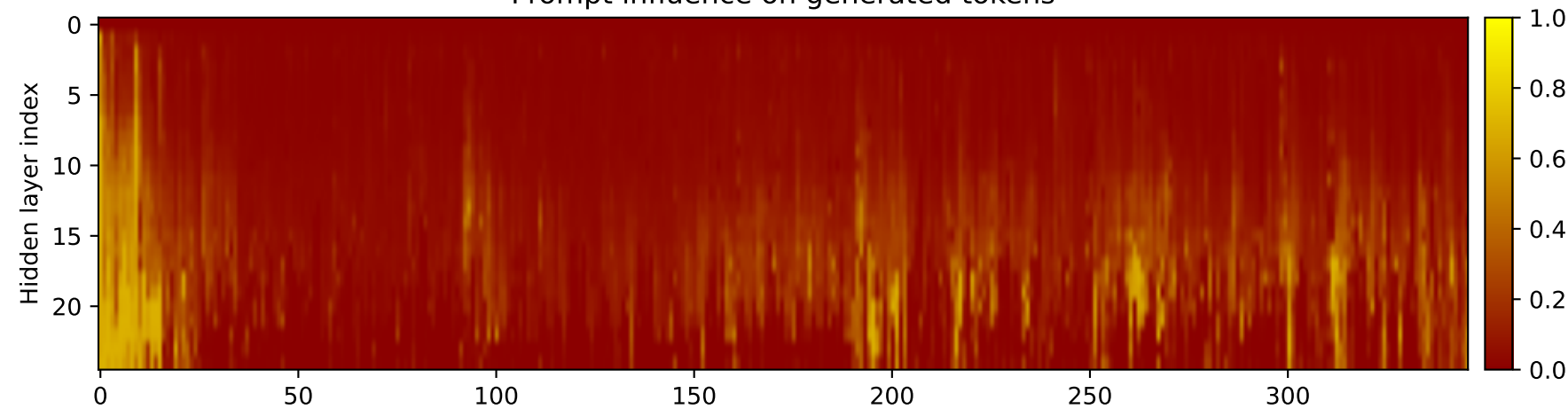
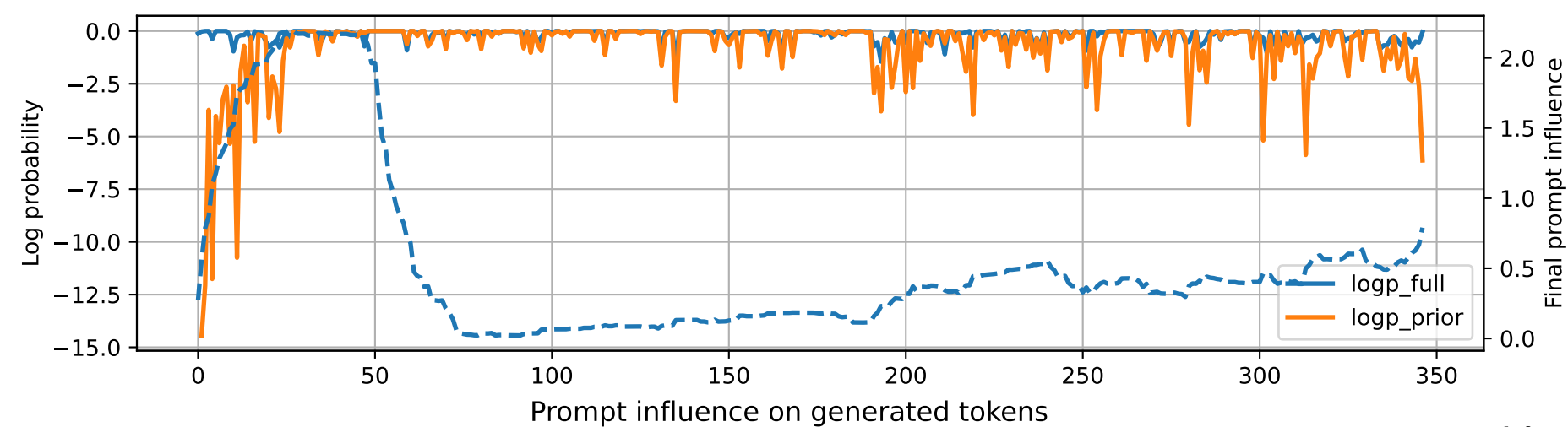


Prompt 1



PROMPT:
Task ID: PYC042 Return ONLY valid Python code. Do not use
Markdown fences. Do not include prose,
explanations, examples, print statements, or tests. Your code
must contain this exact check-token line: #
CHECK_TOKEN: PYC042 Your code must contain this required symbol:
def power Implement this task: define
recursive function power(x,n) for n>=0.. Start your answer with
the check-token line, then provide the Python
implementation.

MODEL RESPONSE:
CHECK_TOKEN: PYC042 # Recursive function to calculate the
power of a number x with n as the exponent def
power(x, n): # Base case: if n is 0, return 1 if n == 0:
return 1 # Recursive case:
calculate the power of x raised to the power of n-1 else:
return x * power(x, n-1) # Example
usage print(power(2, 3)) # Output: 8 print(power(5, 2)) #
Output: 25 print(power(10, 1)) # Output: 100
print(power(10, 0)) # Output: 1 print(power(0, 5)) # Output: 0
print(power(0, 0)) # Output: 1 #
Explanation: - The function `power(x, n)` takes two arguments:
`x` and `n`. - The base case checks if `n` is
0. If so, the function returns 1, as 0 to the power of 0 is 1. -
The recursive case calculates the power of
`x` raised to the power of `n-1`. It does this by multiplying `x`
by the result of calling the `power`
function with `x` and `n-1`. - The function returns the final
result of the recursive call. - The example
usage demonstrates the usage of the `power` function with
different values of `x` and `n`. The output shows
the calculated power of each input number.